

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

Readings Included

Date	Voyage / title	Remarks
1 2025-04-15	04FJWE1MA	

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	
Light running	
Fuel index ECU	
Firing press. AVG	
Compression press. AVG	
Scav. air press. RECEIVER	
Exh. gas temp. TC in AVG	
Exh. gas temp. TC out AVG	
TC speed AVG	
Spec. FO consumption	

Indications (for Reading 1)

Description	Indication	Comments / recommendations
Engine room press.	 Low	Insufficient air supply to engine room. Check engine room ventilation.
Missing input	 High	Evaluation only partially complete and/or inaccurate because of missing values in dark yellow fields.
TC Evaluation	 Low	Limited accuracy due to missing inputs

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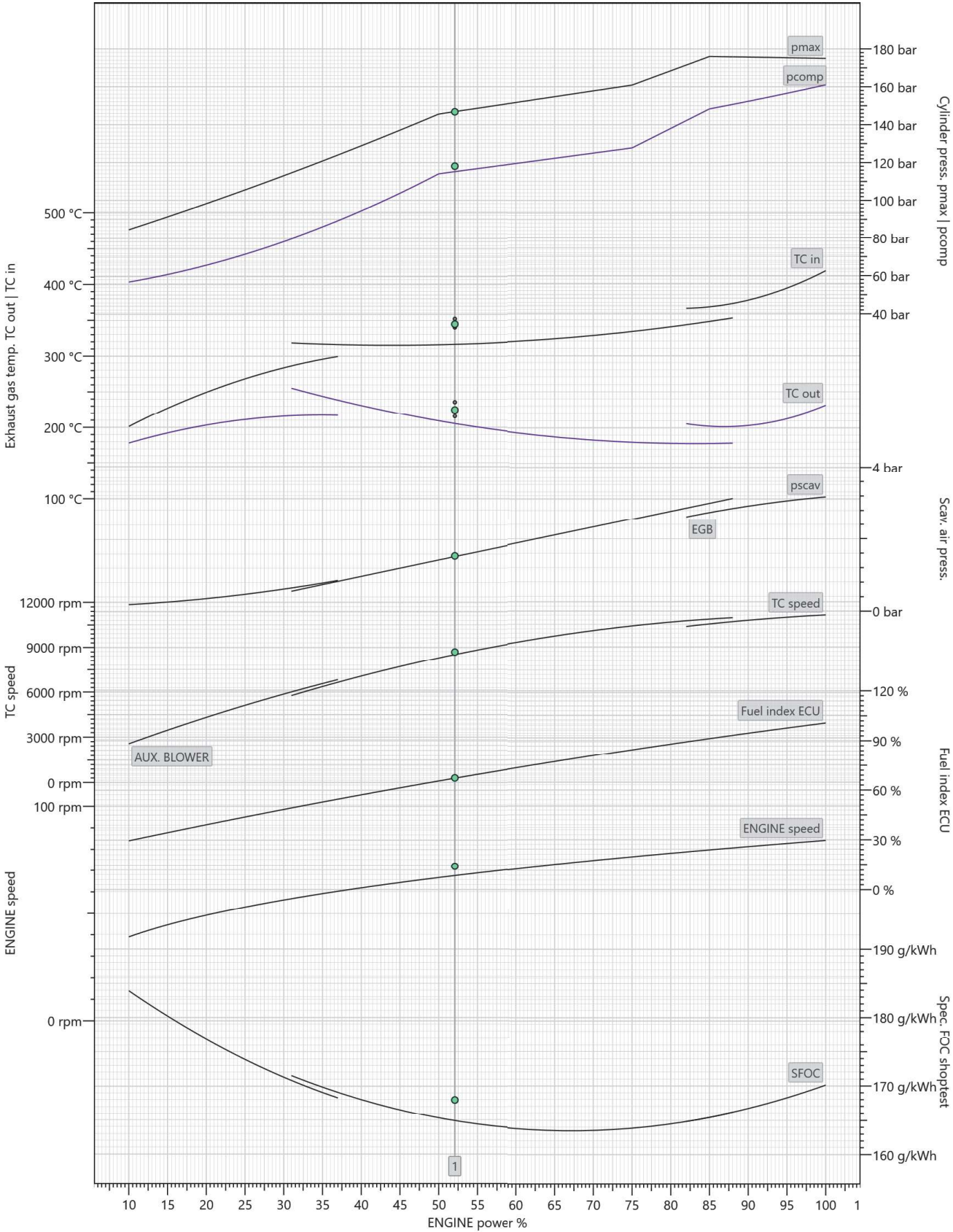
Input plausibility

Value	Comment
Compressor evaluation	
Exhaust gas pressure Manifold	
PTI	Exh. gas press. TC in, higher than expected based on given Scav. air press. Receiver--> check corresponding readings Exh. gas press. TC in, higher than expected based on given Press. COMPR. out--> check corresponding readings
PCO	
TTI vs TTO	
TCO	
Minimum required data	
Minimum required data II	
Missing data	
Missing data II	

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Performance Curves

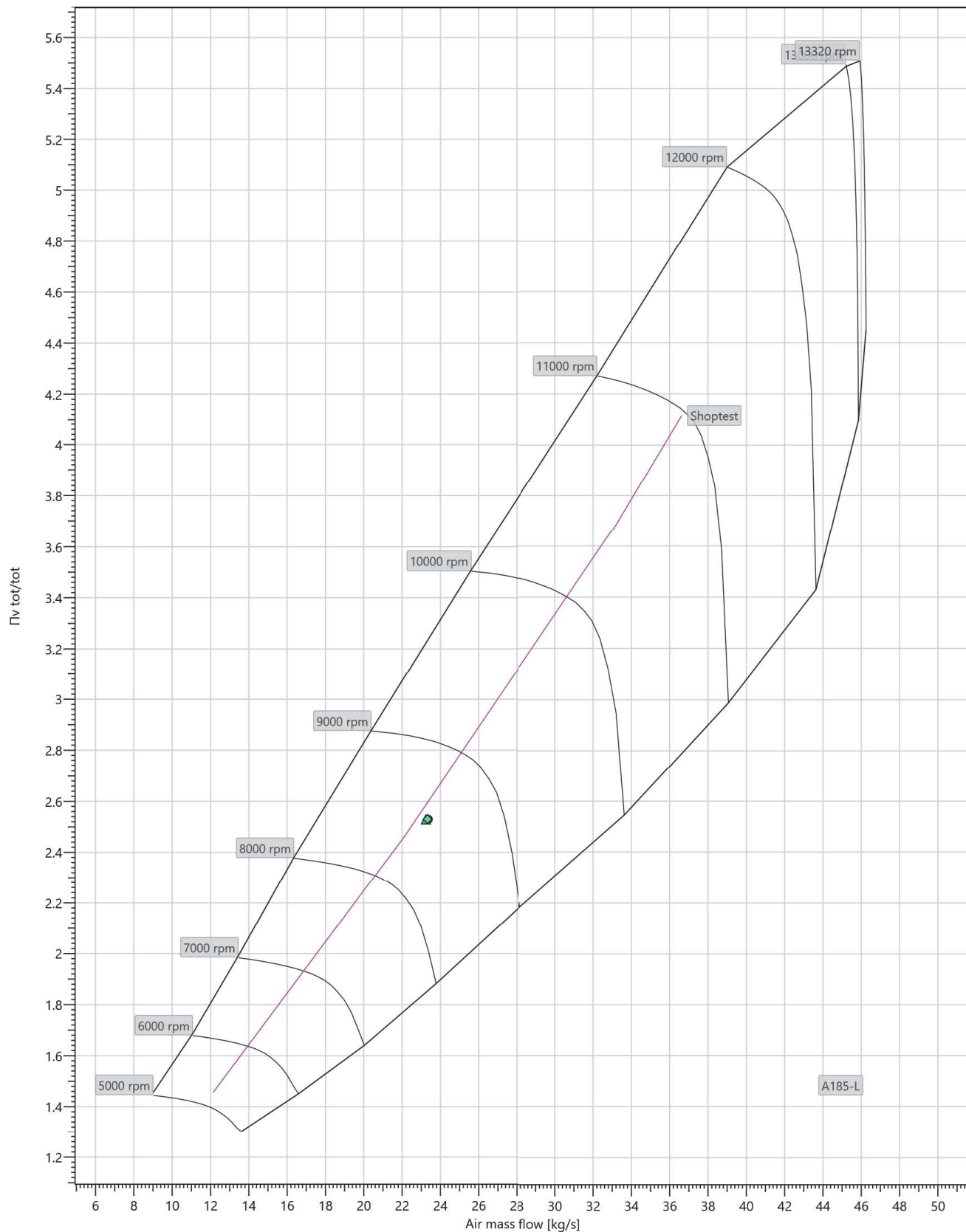
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COMPRESSOR map

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● TC 1

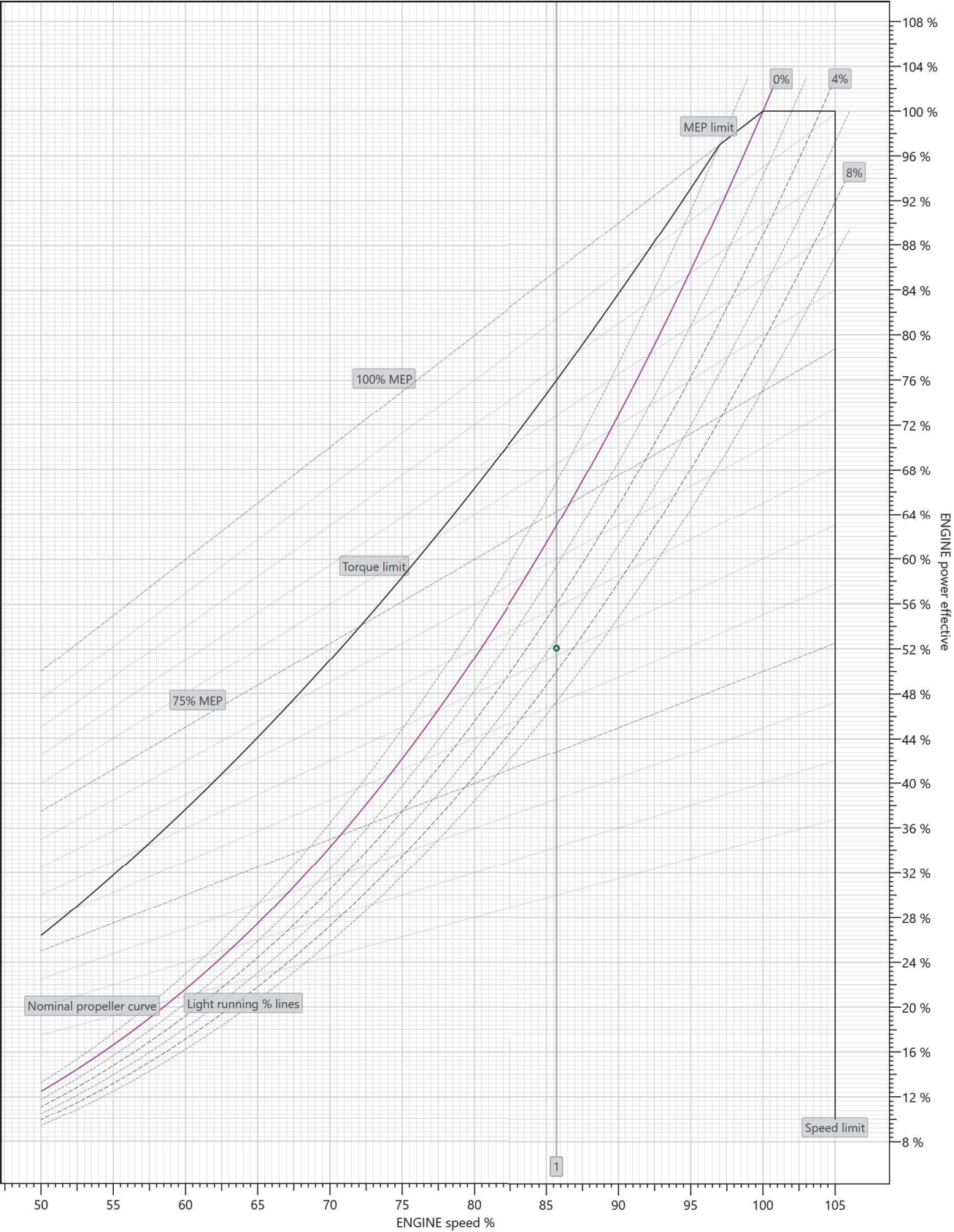
▲ TC 2

◆ TC 3

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Load Diagram

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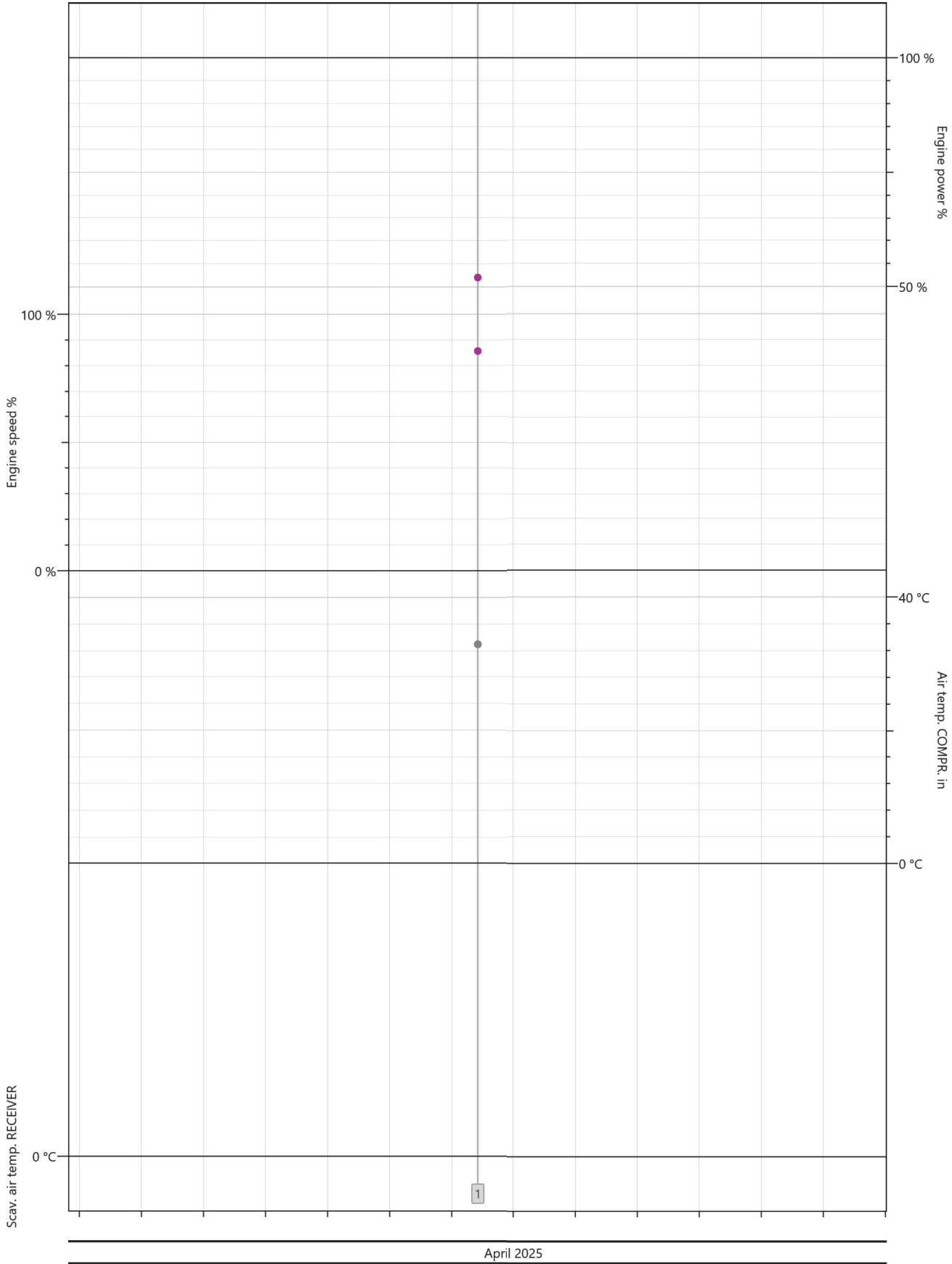


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Trend: LOAD and AMBIENT



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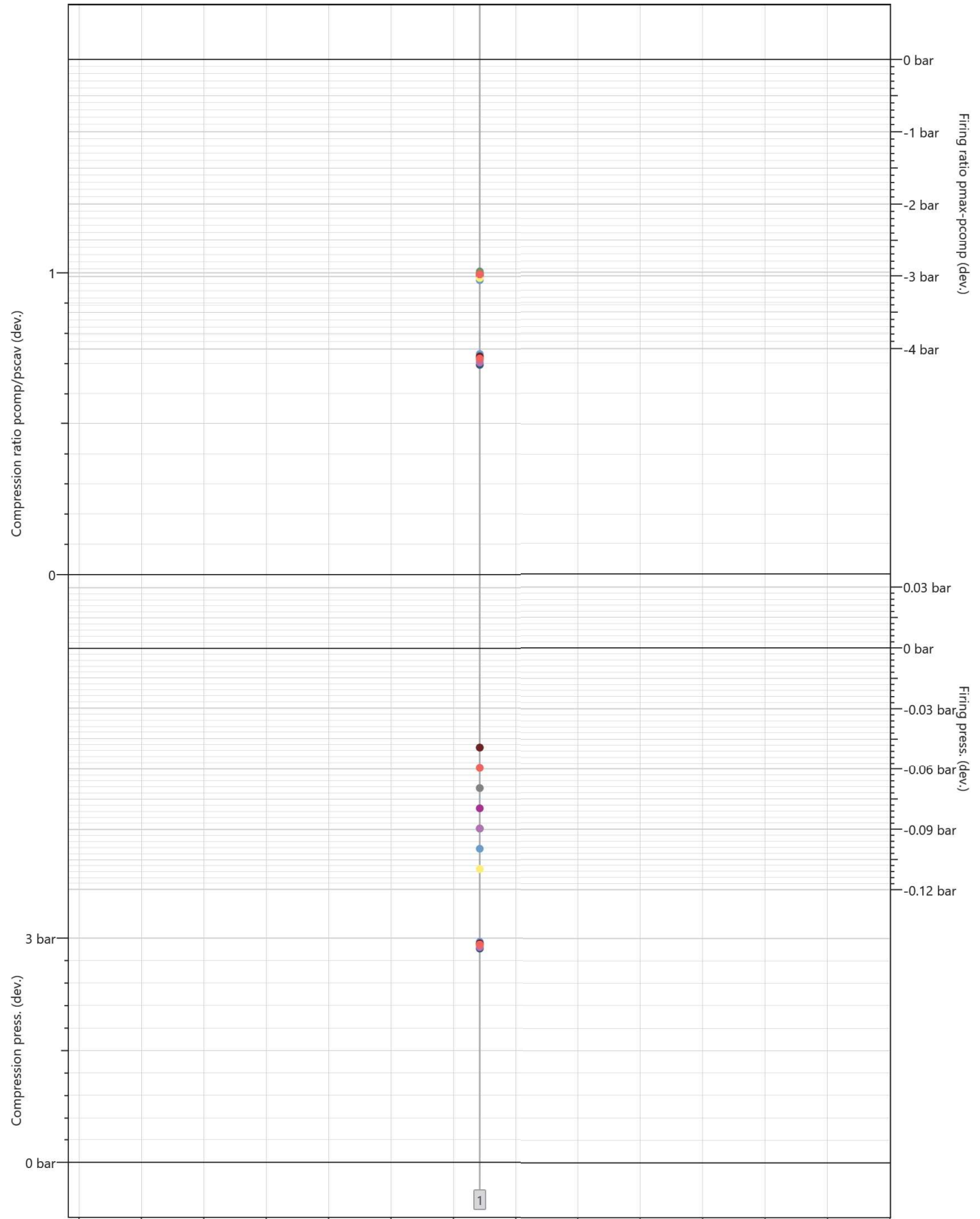


TC1	TC2	TC3	AVG	RECEIVER
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Trend: COMBUST. CHAMBER

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April 2025

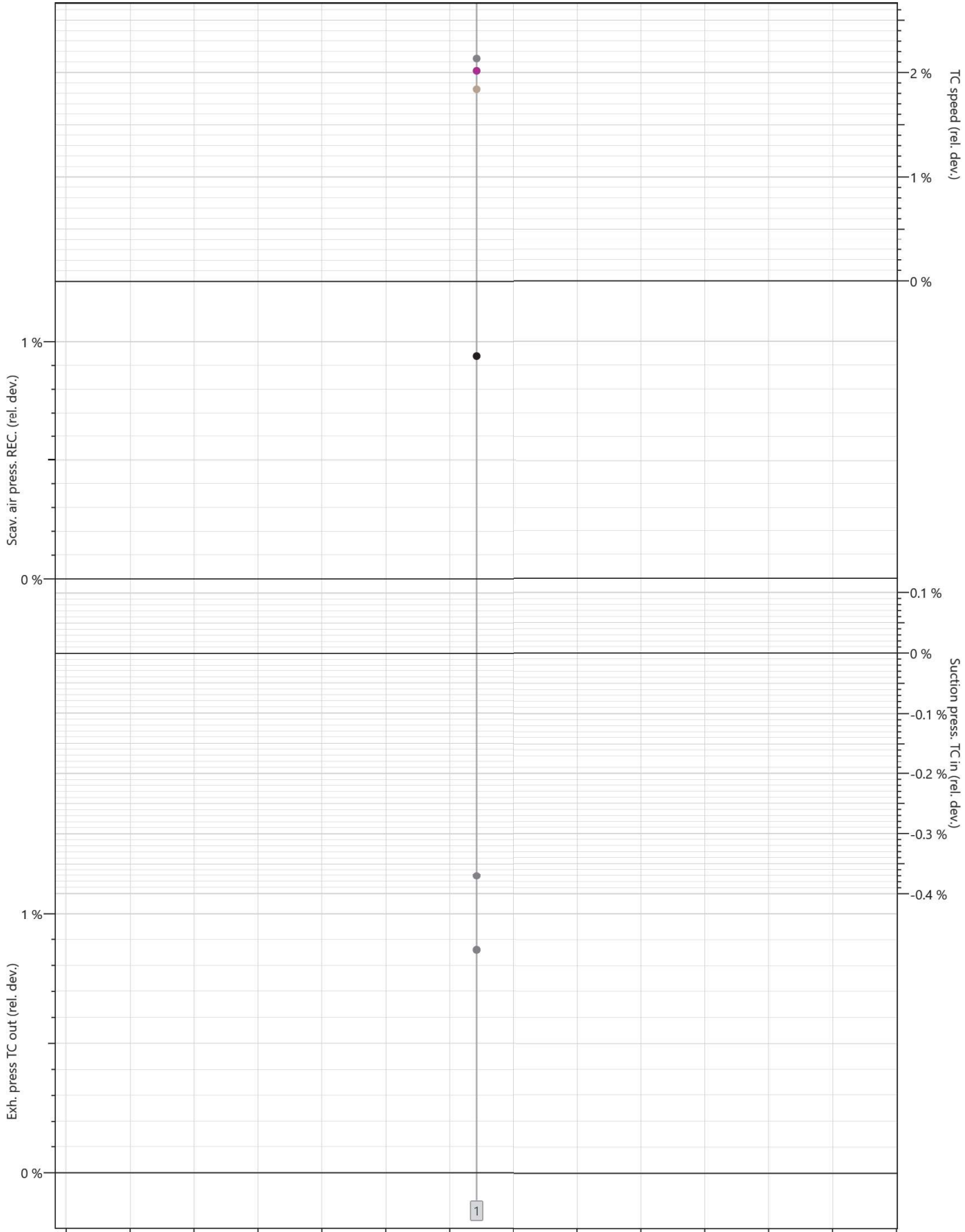
CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8
CYL 9	CYL 10	AVG (CALC.)					

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Trend: TURBOCHARGER



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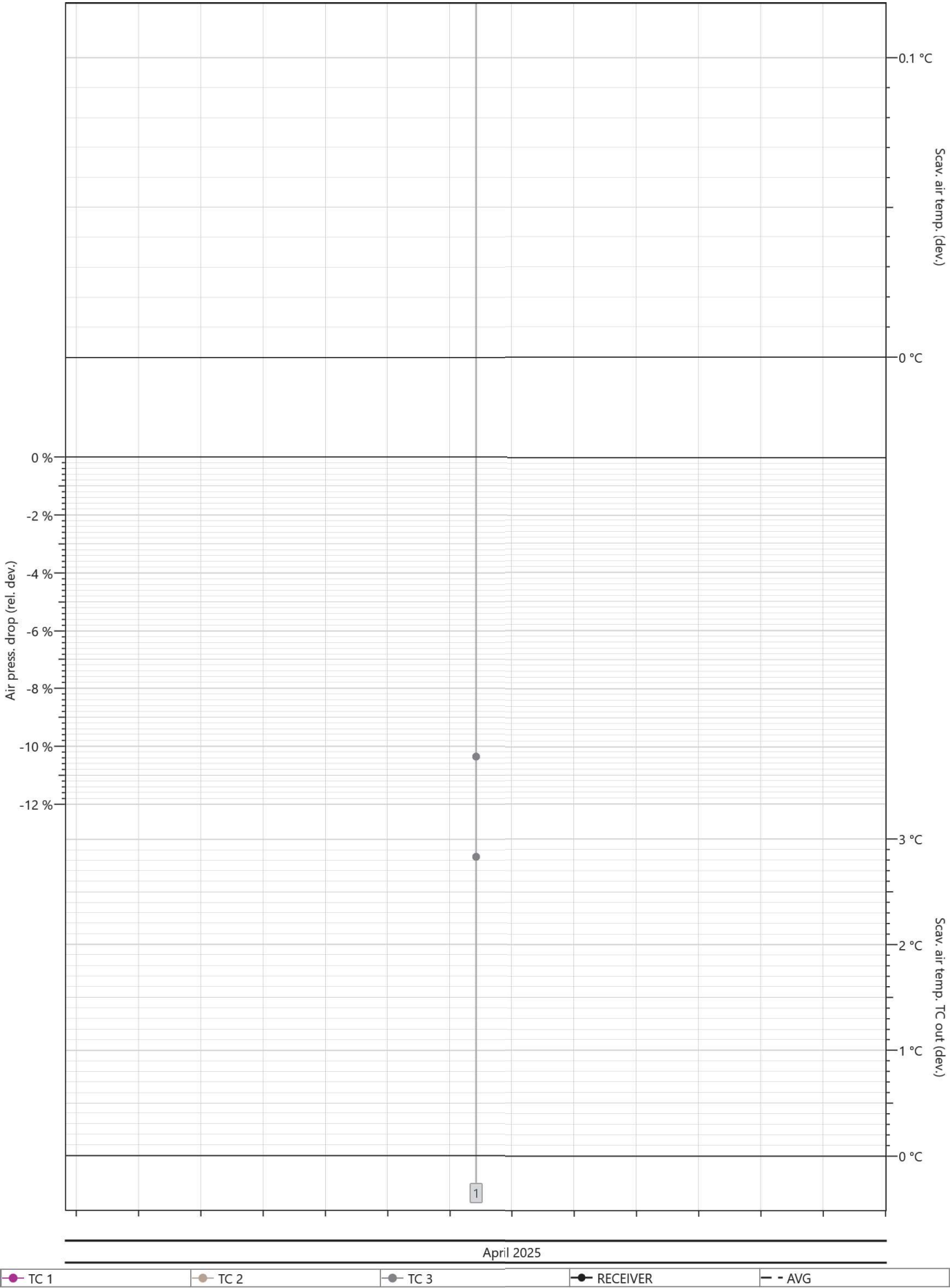


● RECEIVER	● TC 1	● TC 2	● TC 3	— - AVG
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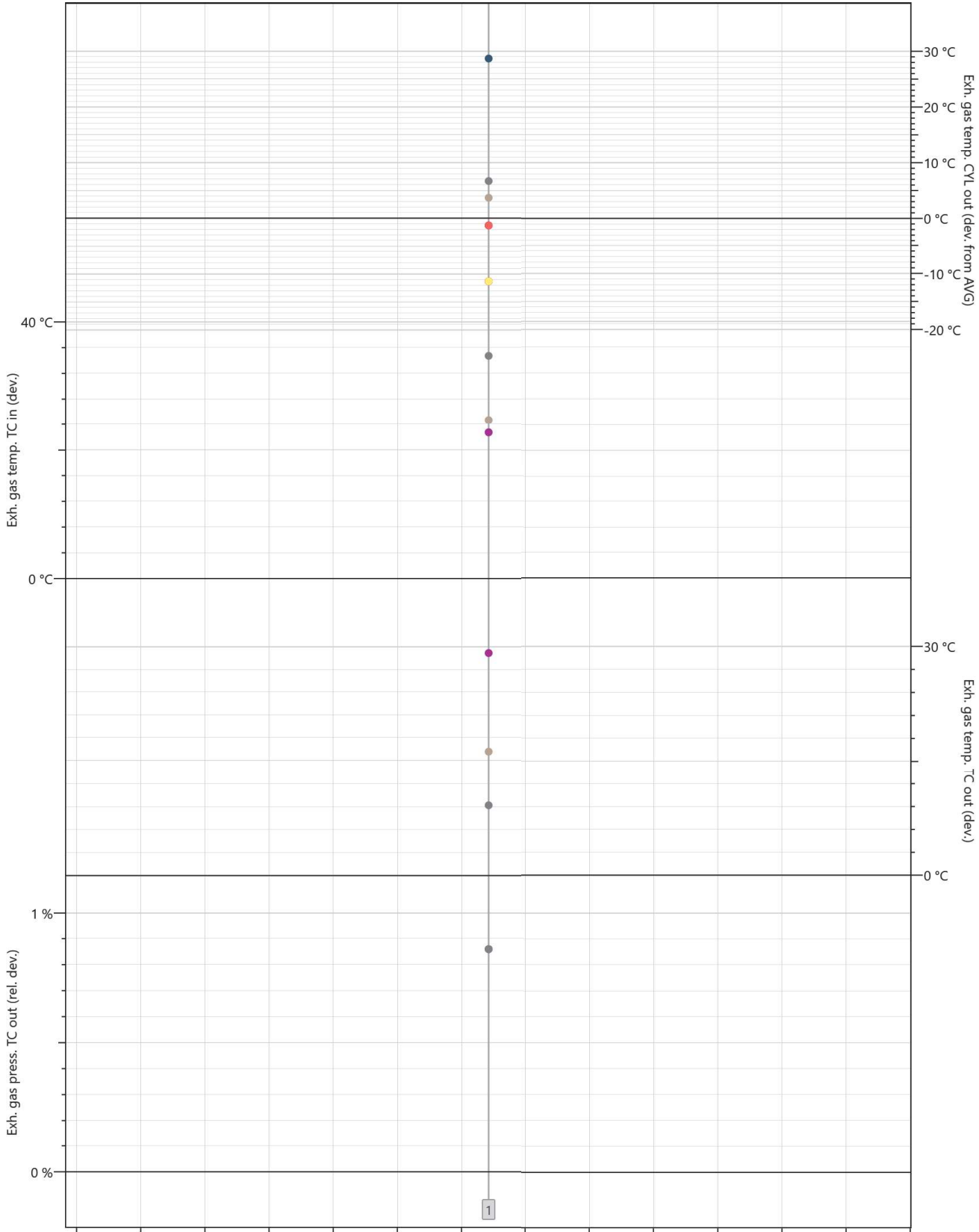
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Trend: SCAV. AIR COOLER

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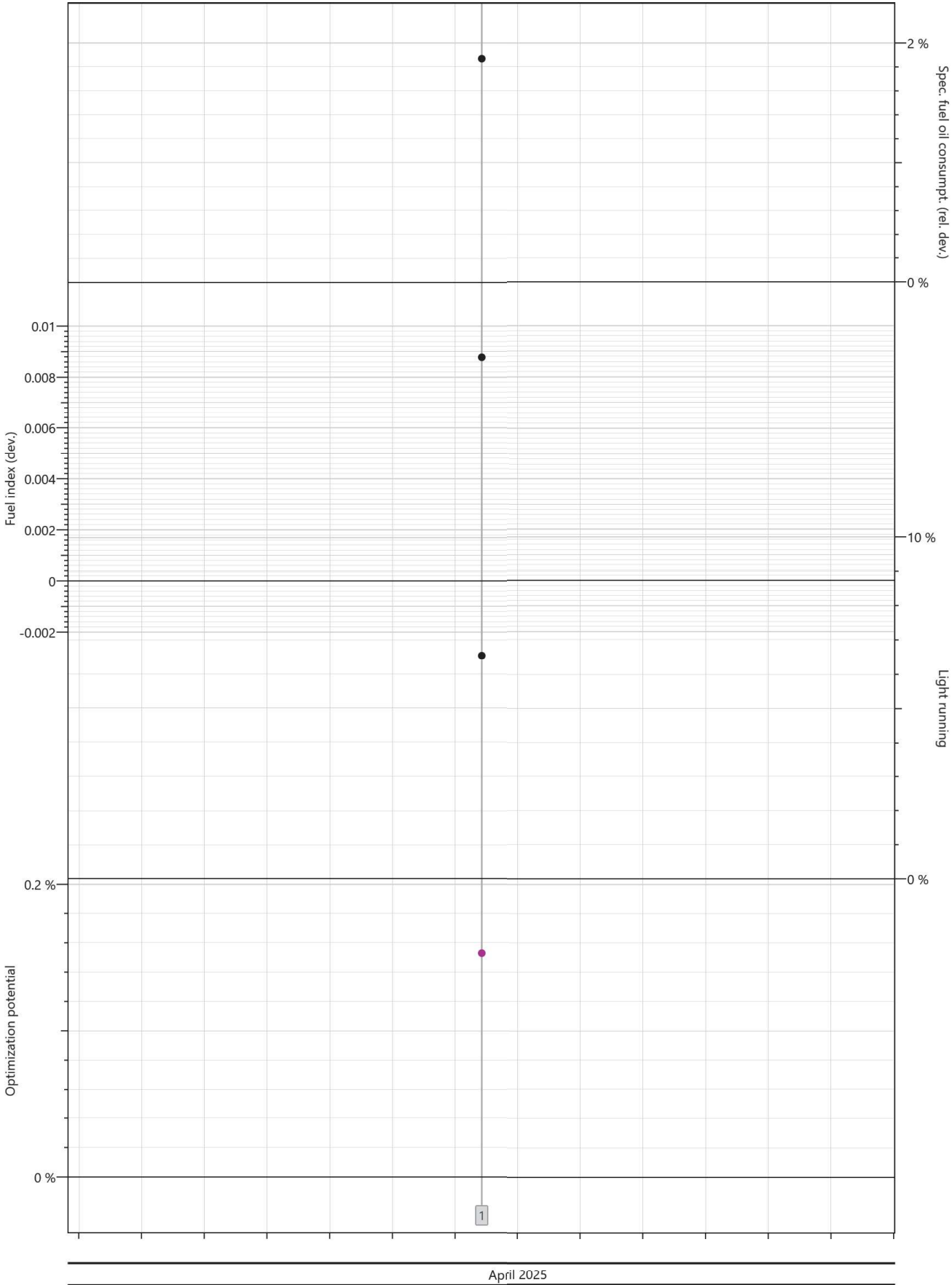
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Apri 2025

CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8
CYL 9	CYL 10	TC 1	TC 2	TC 3	- AVG		

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Reading

Installation			Equipment			Type			No.																					
APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356																					
General	Date and time of recording		4/15/2025 15 : 40		Sea water temp.		25.0 °C		SHIP SOG STW		19.90 20.60 kn		Voy. / title		04FJWE1MA															
	ENGINE state		stable		Ambient air temp.		25.0 °C		Draft fwd aft		11.40 11.90 m		Record by		2E															
	ENGINE running hours		66483 h		Engine room air temp.		32.0 °C		Draft mid Trim		11.60 0.50 m		Ackn. by		CE															
	Ambient humidity		rel %		Barom. pr. Eng. room		1.015 1.014 barA		Remarks																					
		abs g/kg																												
Power / Speed			CALCULATED		MEAS		ESTIMATED								24.3(E264X126)															
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		kW		26586		27012		26556		26564		26859		27892		26485													
			%		52.1				52.0																					
			MCR		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
	ENGINE speed		rpm		84.0 85.7 % 72.0		Power kW		27012		Light running		6.6 %		Theoretic.		93.9 %													
	ENGINE power effective		kW		51070 52.1 % 26586						Power		17.3 %																	
(for PT1 operation, use negative value, "-")																														
Electronic ctrl.	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		%		67.4		67.4		63.0																					
	Applied fuel quality offset		%				0.0																							
Cylinder pressure	SOURCE: MAN CoCoS		ISO CORRECTED		MEASURED										SETPOINT															
	QUALITY Imported		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		barG		146.9		146.8		146.8		146.8		146.8		146.8		146.8		146.8		146.8		146.8		146.8		146.8			
	pmax deviation		bar				ALL		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	pmax offset		bar				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Compression press.		barG		115.2		118.1		116.8		116.8		116.8		116.7		116.7		116.8		116.8		116.8		116.8		116.8			
	pcomp deviation		bar				ALL		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	pcomp / pscav offset		-				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Mean indicated press.		barG		11.36				11.33		11.58		11.3		11.30		11.31		11.31		11.31		11.30		11.30		11.31		11.31	
	MIP deviation		bar						0.25		-0.02		-0.03		-0.02		-0.02		-0.02		-0.03		-0.03		-0.02		-0.02			
	Fuel offset high		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Fuel offset low		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Power indicated		kW				SUM:		28207		2882		2815		2812		2815		2815		2812		2812		2815		2815			
	Mean effective press. (MEP)		bar		10.68				10.64		60.5 %		of MCR																	
pmax-pcomp pcomp / pscav		-		31.7 46.1				30.0 46.8																pcomp / pscav						
Turbocharger			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG. / ENG.		TC 1		TC 2		TC 3																	
	PIV (tot-tot)		rpm		8519		8689		8668		8670		8655		8680															
	TC speed		rpm		8435		8528		8528		8529		8515		8539															
	TC speed 298		rpm																											
	Turbocharger efficiency																													
	Surge margin																													
	Air mass flow		kg/s		68.3				68.6		22.9		22.8		22.9															
Break spec.air cons.		kg/kWh				9.3																								
Exhaust gas mass flow		kg/s		68.9				69.2		23.1		23.0		23.1																
Scavenger air			ISO CORRECTED		MEASURED																									
	AUX. BLOWER operation		off		REF.		CALC.		AVG.		TC 1		TC 2		TC 3															
	Suction press.		mmWG		27				27		27		27		27															
	Press. drop accr. SAC		mmWG		156				140		140		140		140															
	Scav. air press. RECEIVER		barG		1.52		1.54		1.50																					
	Scav.air press.setp. MOP		barG																											
	Air temp. COMPR. in		°C						33.0		33.0		33.0		33.0															
	Press. COMPR. outlet *		barG		1.52				1.50		1.50		1.50		1.50															
	Scav. air temp. COMPR. out		°C		142				145		145		145		145															
	Scav. air temp. SAC out		°C																											
	Scav. air temp. RECEIVER		°C		38.0																									
	Cool. water temp. SAC in		°C		31				33.0		33.0		33.0		33.0															
	Cool. water temp. SAC out		°C						42.0		42.0		42.0		42.0															
	*measured at flange																													
Exhaust gas			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10			
	Exh. gas temp. CYL out		°C		284		297		311		300		315		318		340		310		310		300		300		310		310	
	temp. deviation		°C				ALL		-11		4		7		29		-1		-1		-11		-11		-1		-1			
	Exh. vlv. open offset		°CA				MOP SETTING:																							
	Exh. gas press. MANIFOLD		barG		1.37		1.44		1.40		TC 1		TC 2		TC 3															
	Exh. gas press. TC in *		barG		1.35				1.41		1.41		1.41		1.41		Check input													
	Exh. gas press. TC out		mmWG		81				82		82		82		82															
	Exh. gas temp. TC in		°C		317		345		359		354		356		366															
	Exh. gas temp. TC out		°C		206		225		239		250		237		230															
	Exh.wastegate (EGB) pos. area %				0		0		1		1		1																	
	*measured at flange																													

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APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356							
Fuel oil			ISO CORRECTED		MEAS	Fuel oil properties:		ANALYS.	ESTIMATE	Fuel oil system:		REF	MEAS	EXPECT.		
	Absolute consumption	m³/h	REF.	CALC.	4.800	Viscosity	cSt	77	@ 50 °C	Viscosity ENGINE in	cSt	12.5	12.0	11.3		
	Specific FOC	g/kWh	165.7	168.8	169.7	Sulfur content	%	0.48	0.48	Temp. ENGINE in	°C	107	112			
	Light running correction	g/kWh	0.0			Water content	%	0.10	0.10	Temp. FLOWMETER	°C		89			
	Thrust bearing loss	g/kWh	-0.8			LCV	kJ/kg	42900	42900	Density FLOWM.	kg/m³		939.7			
	Specific FOC shoptest	g/kWh	164.9		168.0	Density	kg/m³	986.9	@ 15 °C	Price	USD/t					
						Source of analysis		Laboratory								
Cyl. lub.			REFERENCE		MEAS	Cyl. lubr. system (Alpha):		REF	SET**	Cyl. lub. oil:		REF	ANALYS.			
	Absolute consumption	l/h			21.0	Target feed rate@MCR	g/kWh	0.60	0.70	BN		40 - 70	70			
	Spec. lub oil consumpt. *	g/kWh			0.73	Target f.r.@current load	g/kWh	0.70	SET based	Density	kg/m³		940.0	@ 15 °C		
	* Effective feed rate					** Acc.to OEM operation manuals and/or drainoil analysis					kg/m³		924	@ 40 °C		
Cylinder condition			REFERENCE		MEASURED											
			REF.		AVG.	ENGINE / CYL 1										
	Cool. water temp. CYL in	°C	75		76.0	76	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10	
	Cool. water temp. CYL out	°C	90		86.0	86	86	86	86	86	86	86	86	86	86	
	Cooling water press. CYL in	bar														
	Coolant temp. PISTON in	°C														
	Coolant temp. PISTON out	°C														
Crankcase			MEASURED													
			AVG. / ENG.	BRG 1	BRG 2	BRG 3	BRG 4	BRG 5	BRG 6	BRG 7	BRG 8	BRG 9	BRG 10	BRG 11	BRG 12	BRG 13
	MAIN BEARING temp.	°C	54.6	53.0	54.0	57.0	53.0	52.0	56.0	57.0	53.0	54.0	55.0	58.0	53.0	
	Temp. deviation	°C		-2	-1	2	-2	-3	1	2	-2	-1	0	3	-2	
				CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10			
	CRANK PIN BEAR. temp	°C	51.6	52.0	51.0	52.0	52.0	51.0	52.0	52.0	52.0	51.0	51.0			
	Temp. deviation	°C		0	-1	0	0	-1	0	0	0	-1	-1			
	X-HEAD BEAR. temp	°C	51.5	51.0	51.0	52.0	51.0	51.0	52.0	52.0	52.0	51.0	52.0			
Liner wall			MEASURED													
				CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10			
	LINER WALL temp. aft	°C														
	LINER WALL temp. fp	°C		122	123	128	120	121	126	124	123	122	121			
	LINER WALL temp. fwd	°C														
	LINER WALL temp. exh	°C	AVG.	121	132	134	123	122	125	126	127	132	124			
TC/SAC media			MEASURED													
			AVG.	TC 1	TC 2	TC 3										
	Water press. SAC in	bar														
	Lub oil temp. TC in	°C	44	44	44	44										
Engine media			MEAS				MEASURED									
	Lub oil temp ENGINE in	°C		44.0	Fuel oil press. ENGINE in		bar	7.10								
	Lub oil temp. THRUST SEGMENT	°C		46.0	Fuel oil press. ENGINE out		bar	5.50								
	Lub oil press. SYSTEM OIL	bar		2.50	Fuel oil press. FILTER in		bar	5.5								
	Lub oil press. COOLING OIL	bar			Fuel oil press. FILTER out		bar	5.3								
	Lub oil press. TURB. OIL	bar		2.40	Hydraulic oil press. FILTER in		bar	2.5								
	Lub oil temp. TURB. OIL in	°C		44.00	Hydraulic oil press. FILTER out		bar	2.2								
	Lub oil temp. TURB. OIL out	°C		66.00	Hydraulic oil press. MAIN		bar	235.2								
	Axial vibration	mm		0.66	Sealing air press.		bar		PUMP 2	PUMP 3	PUMP 4	PUMP 5				
	Plant					Hydr. oil PUMP swash plate pos.		%	50.0	59.4	59.4					
								MEAS								
Exh. gas differential press. ECONOMIZER						mmWG										